Knowledge Organiser
Year 7: Cells – Exercise and Respiration

What is respiration?
- Respiration is a chemical reaction which occurs in all living cells
- It releases energy from glucouse
- Respiration happens in the mitochondria

Why is it important?
- Respiration is not breathing, it is a chemical reaction which releases energy
- Without it cells would not have the energy they need to do their jobs

Chemical equation for **aerobic respiration:**

GLUCOSE + OXYGEN ⟶ CARBON DIOXIDE + WATER

How does aerobic respiration work?
- Mitochondria react together glucose and oxygen to form carbon dioxide and water
- Energy is released during this reaction which cells can use
- Aerobic respiration only works if there is plenty of oxygen

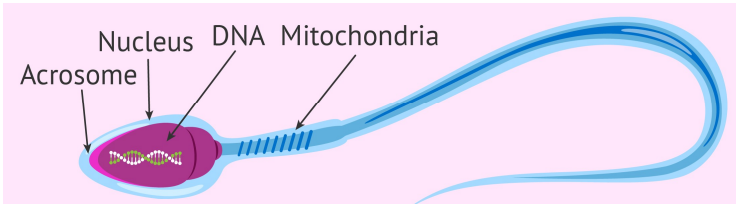
What is anaerobic respiration?
- Anaerobic respiration happens when there is not enough oxygen getting to the cell
- A harmful by-product called lactic acid is formed
- Anaerobic respiration is **less efficient** than aerobic respiration – it releases less energy
- It happens when we exercise vigorously



Comparing aerobic and anaerobic respiration:

	Aerobic	Anaerobic
Releases energy from glucouse	✓	✓
Can happen in all cells	✓	✓
Produces only carbon dioxide and water	✓	
Produces lactic acid		✓
Happens in plentiful oxygen	✓	
Happens without oxygen		✓

Adaptations for respiration?
- Some cells are **adapted** for plentiful respiration
- Sperm cells are very active during their short lives, they must swim quickly to meet the egg cell and need lots of energy



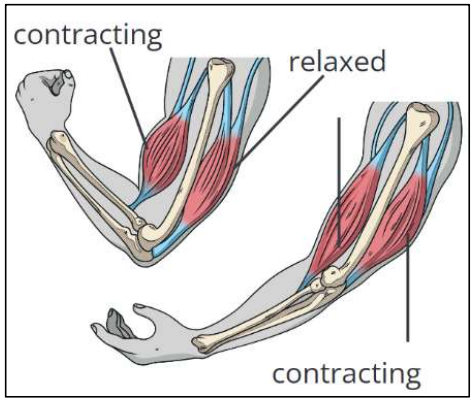
- See how the sperm cell has lots of **densely packed mitochondria**
- This is the organelle responsible for respiration
- The sperm cell needs lots of mitochondria so it can release plenty of energy from respiration for swimming



How do muscles work?
- Muscles are bundles of fibres which can stretch and store energy
- They transfer the chemical energy from glucouse into kinetic (movement)
- Muscles can only contract or relax, this means they can pull and not push

Antagonistic pairs of muscles?
- A pair of muscles that work together to move are called **antagonistic pairs**
- When one of the muscles **contracts** the other **relaxes**
- This pair of movements pulls your bones in one direction or the other

This diagram shows how the bicep and tricep work together to make your arm move



RAG Yourself!

Learning Outcomes;	RAG
Describe why respiration is important	
Give the word equation for aerobic respiration	
Compare aerobic and anaerobic respiration	
Describe how a sperm cell is adapted for respiration	
Explain how antagonistic pairs of muscles work	
Explain why heart rate increases during exercise	



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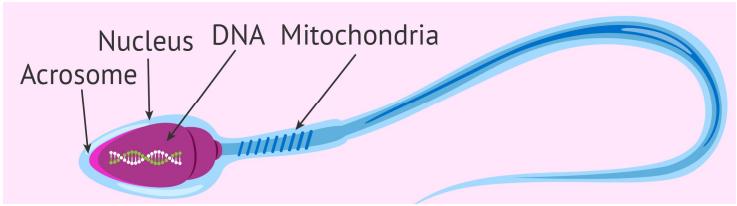
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Antagonistic pairs of muscles?

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